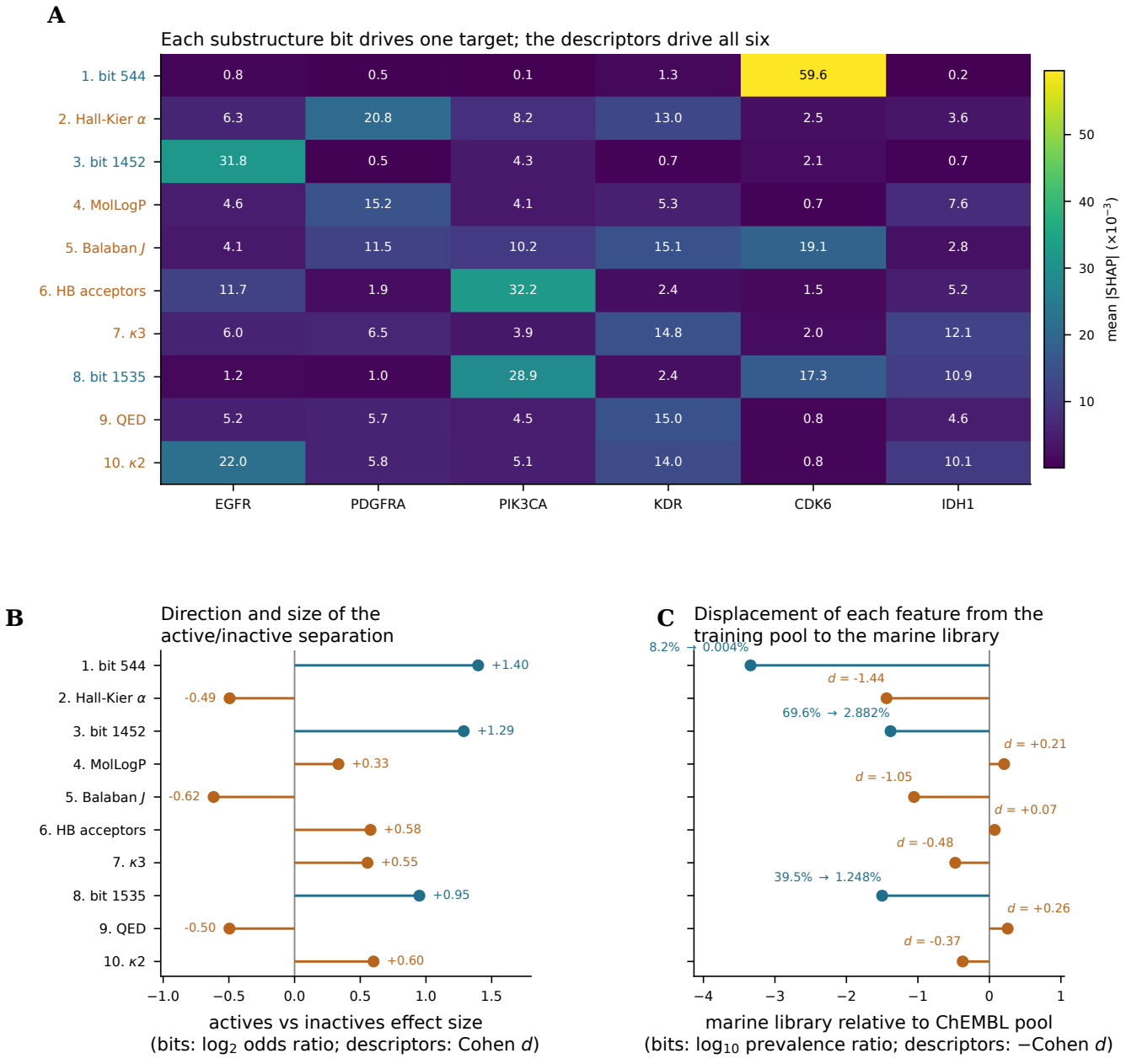
A
Each substructure bit drives one target; the descriptors drive all six
1. bit 544
2. Hall-Kier α
3. bit 1452
4. MolLogP
5. Balaban J
6. HB acceptors
7. κ3
8. bit 1535
9. QED
10. κ2
0.8 0.5 0.1 1.3 59.6 0.2
6.3 20.8 8.2 13.0 2.5 3.6
31.8 0.5 4.3 0.7 2.1 0.7
4.6 15.2 4.1 5.3 0.7 7.6
4.1 11.5 10.2 15.1 19.1 2.8
11.7 1.9 32.2 2.4 1.5 5.2
6.0 6.5 3.9 14.8 2.0 12.1
1.2 1.0 28.9 2.4 17.3 10.9
5.2 5.7 4.5 15.0 0.8 4.6
22.0 5.8 5.1 14.0 0.8 10.1
EGFR
PDGFRA
PIK3CA
KDR
CDK6
IDH1
mean |SHAP| (×10⁻³)
50
40
30
20
10
B
Direction and size of the active/inactive separation
1. bit 544
2. Hall-Kier α
3. bit 1452
4. MolLogP
5. Balaban J
6. HB acceptors
7. κ3
8. bit 1535
9. QED
10. κ2
+1.40
-0.49
+1.29
+0.33
-0.62
+0.58
+0.55
+0.95
-0.50
+0.60
−1.0 −0.5 0.0 0.5 1.0 1.5
actives vs inactives effect size
(bits: log2 odds ratio; descriptors: Cohen d)
C
Displacement of each feature from the training pool to the marine library
8.2% → 0.004%
d = -1.44
69.6% → 2.882%
d = +0.21
d = -1.05
d = +0.07
d = -0.48
39.5% → 1.248%
d = +0.26
d = -0.37
−4 −3 −2 −1 0 1
marine library relative to ChEMBL pool
(bits: log10 prevalence ratio; descriptors: −Cohen d)